Exploratory Behavioral Analysis

This notebook explores behavioral differences between annual members and casual riders using engineered features.

The objective is to identify:

- riding behavior patterns
- commuting versus leisure usage
- temporal demand differences
- potential membership conversion opportunities

The analysis focuses on generating business-relevant insights rather than descriptive statistics alone.

Dataset Overview

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5541823 entries, 0 to 5541822
Data columns (total 21 columns):
 #   Column              Dtype
---  ------              -----
 0   ride_id             object
 1   rideable_type       category
 2   started_at          datetime64[ns]
 3   ended_at            datetime64[ns]
 4   start_station_name  category
 5   start_station_id    object
 6   end_station_name    category
 7   end_station_id      object
 8   start_lat           float64
 9   start_lng           float64
 10  end_lat             float64
 11  end_lng             float64
 12  member_casual       category
 13  ride_length_min     float64
 14  day_of_week         object
 15  month               object
 16  hour                int32
 17  date                object
 18  is_weekend          bool
 19  season              object
 20  time_period         object
dtypes: bool(1), category(4), datetime64[ns](2), float64(5), int32(1), object(8)
memory usage: 692.5+ MB
```

```
is_round_trip
False    4889507
True      652316
Name: count, dtype: int64
```

```
member_casual
casual    0.154716
member    0.097489
Name: is_round_trip, dtype: float64
```

	count	mean	std	min	25%	50%	75%	max
member_casual								
casual	1957976.0	23.347497	82.935809	1.0	6.77	11.81	21.58	1574.9
member	3583847.0	12.690988	33.202348	1.0	5.22	8.75	14.74	1559.9

```
member_casual
member    0.646691
casual    0.353309
Name: proportion, dtype: float64
```

is_weekend	False	True
member_casual		
casual	1232778	725198
member	2757277	826570

```
member_casual  is_weekend
casual         False         62.96
               True          37.04
member         False         76.94
               True          23.06
Name: proportion, dtype: float64
```

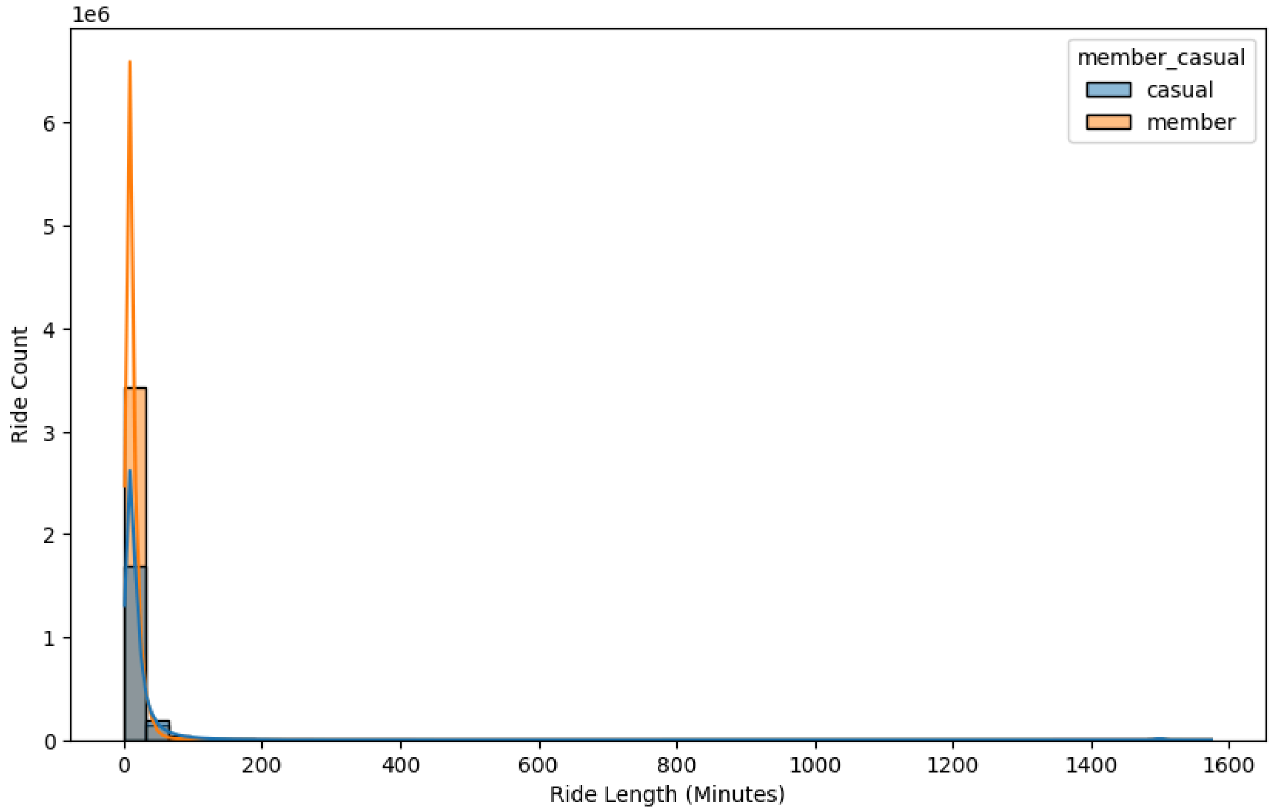
season	fall	spring	summer	winter
member_casual				
casual	565039	388562	913073	91302
member	1107371	846382	1253415	376679

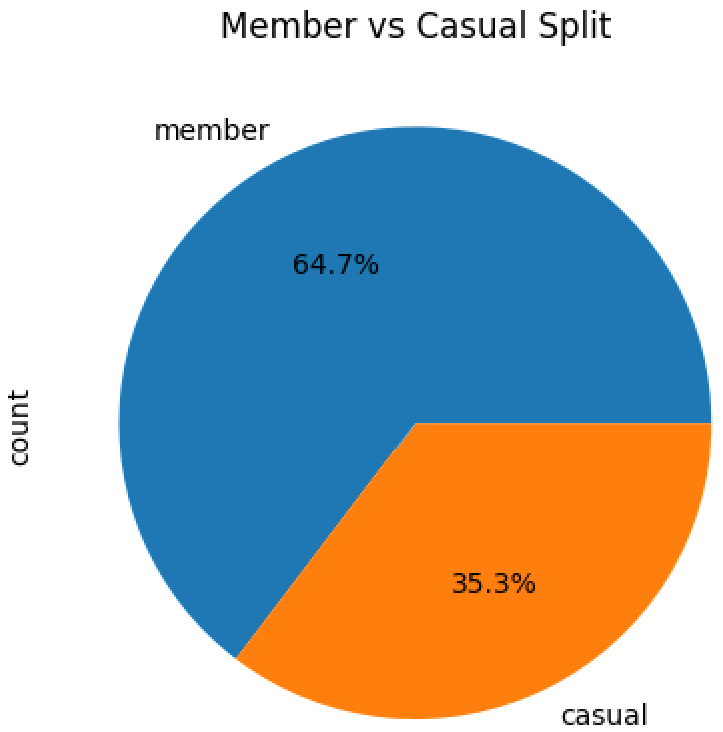
```
member_casual  member_casual  start_station_name
casual         casual         Navy Pier                              32221
                              DuSable Lake Shore Dr & Monroe St      31175
                              Michigan Ave & Oak St                  22325
                              DuSable Lake Shore Dr & North Blvd     19319
                              Streeter Dr & Grand Ave                18982
member         member         Kingsbury St & Kinzie St               23034
                              Canal St & Madison St                  22291
                              Clinton St & Washington Blvd           21952
                              Clinton St & Madison St                20448
                              State St & Chicago Ave                 19692
dtype: int64
```

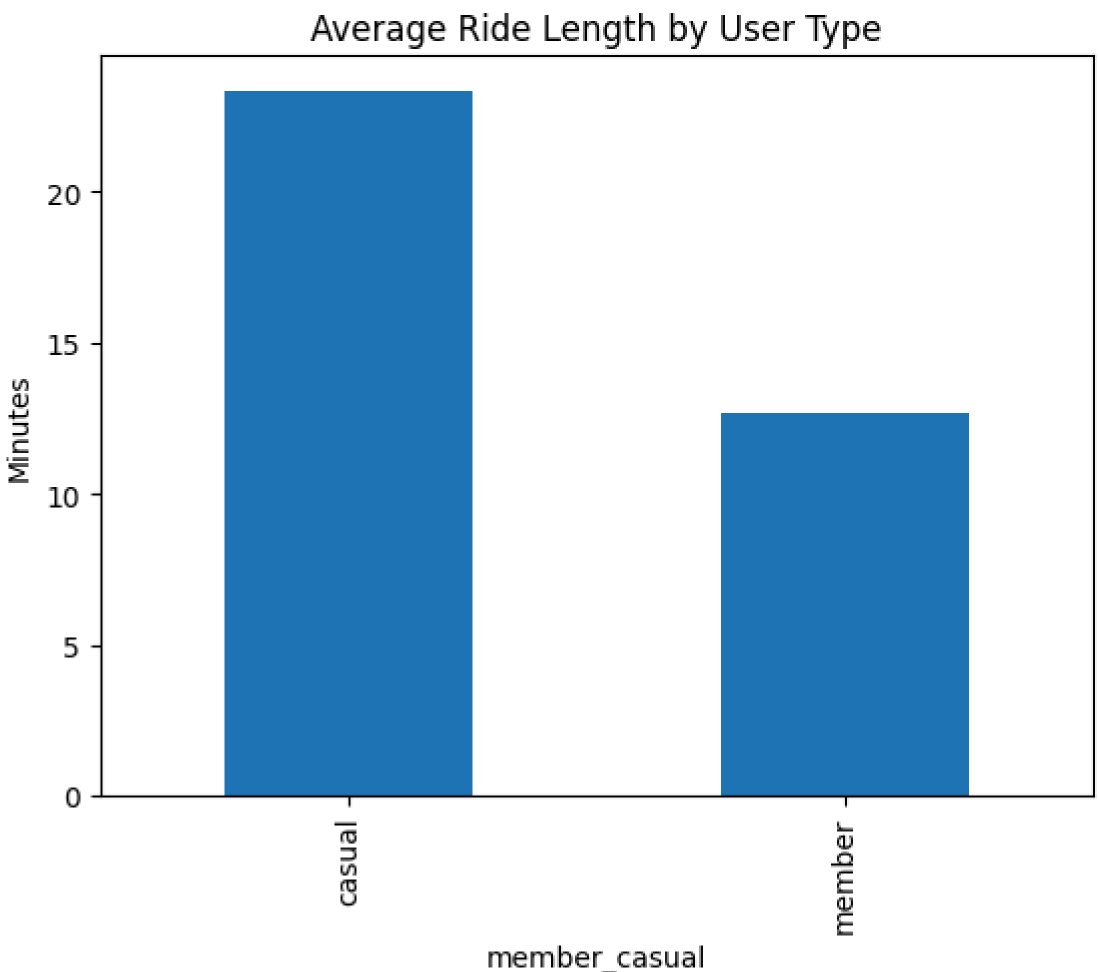

```
member_casual
member    0.646691
casual    0.353309
Name: proportion, dtype: float64
```

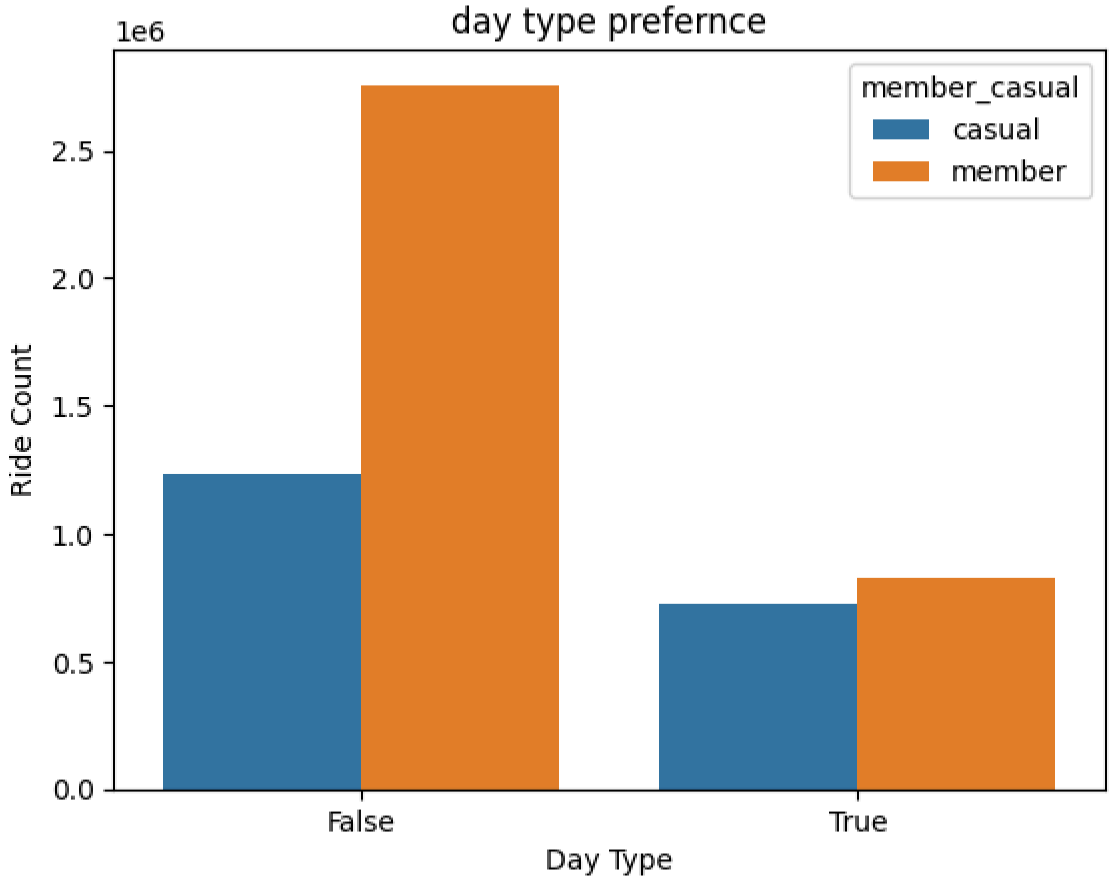
		count	mean	std	min	25%	50%	75%	max
member_casual	rideable_type								
casual	classic_bike	662953.0	39.986820	139.308560	1.0	9.08	16.40	31.38	1574.90
	electric_bike	1295023.0	14.829433	15.824544	1.0	6.05	10.18	17.68	479.97
member	classic_bike	1273360.0	15.094434	53.002495	1.0	5.38	9.26	16.45	1559.90
	electric_bike	2310487.0	11.366396	12.520622	1.0	5.14	8.51	13.96	480.48

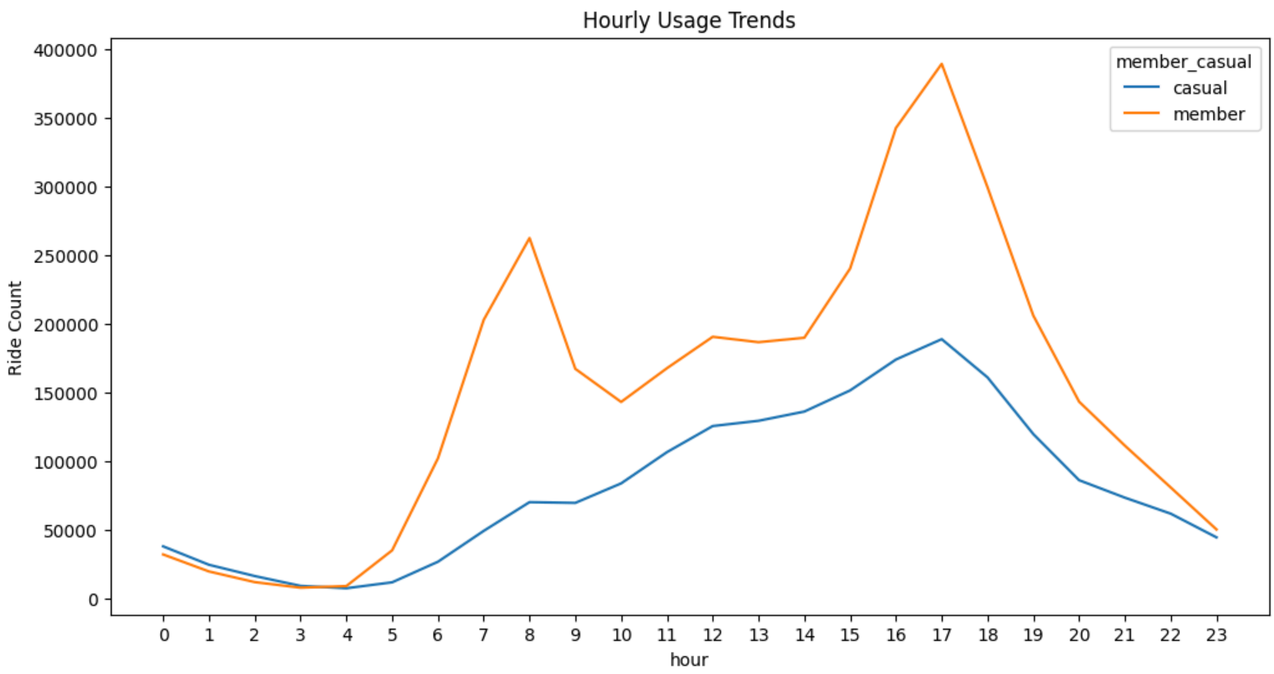
```
member_casual
casual    23.347497
member    12.690988
Name: ride_length_min, dtype: float64
```

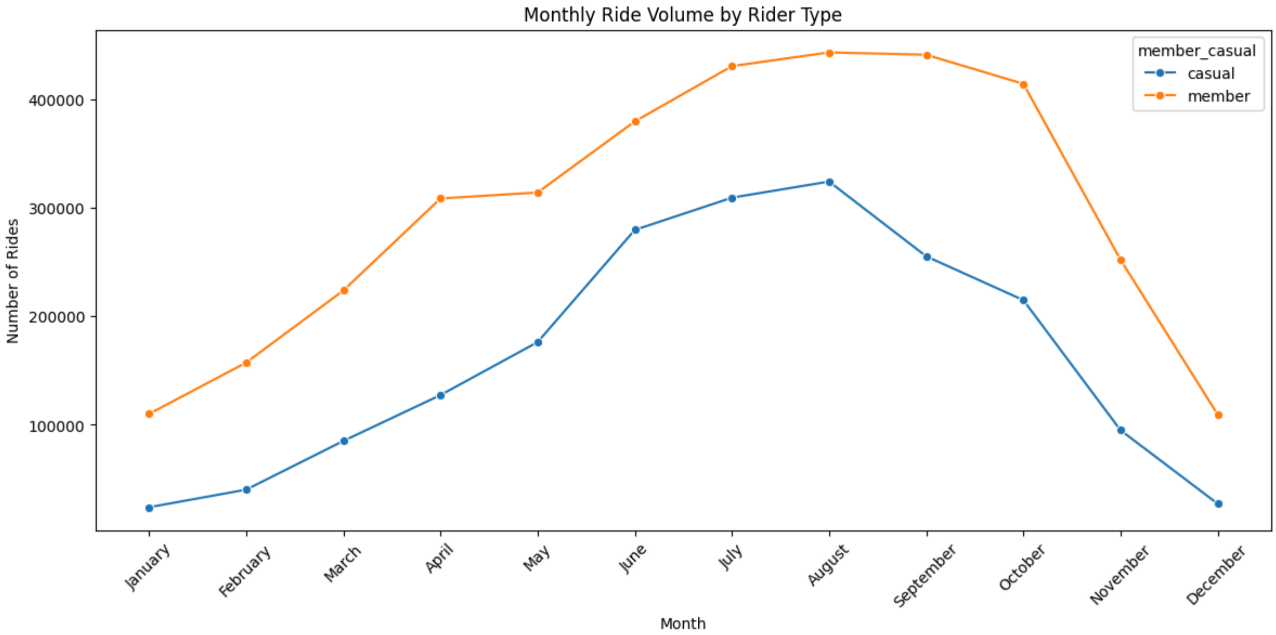


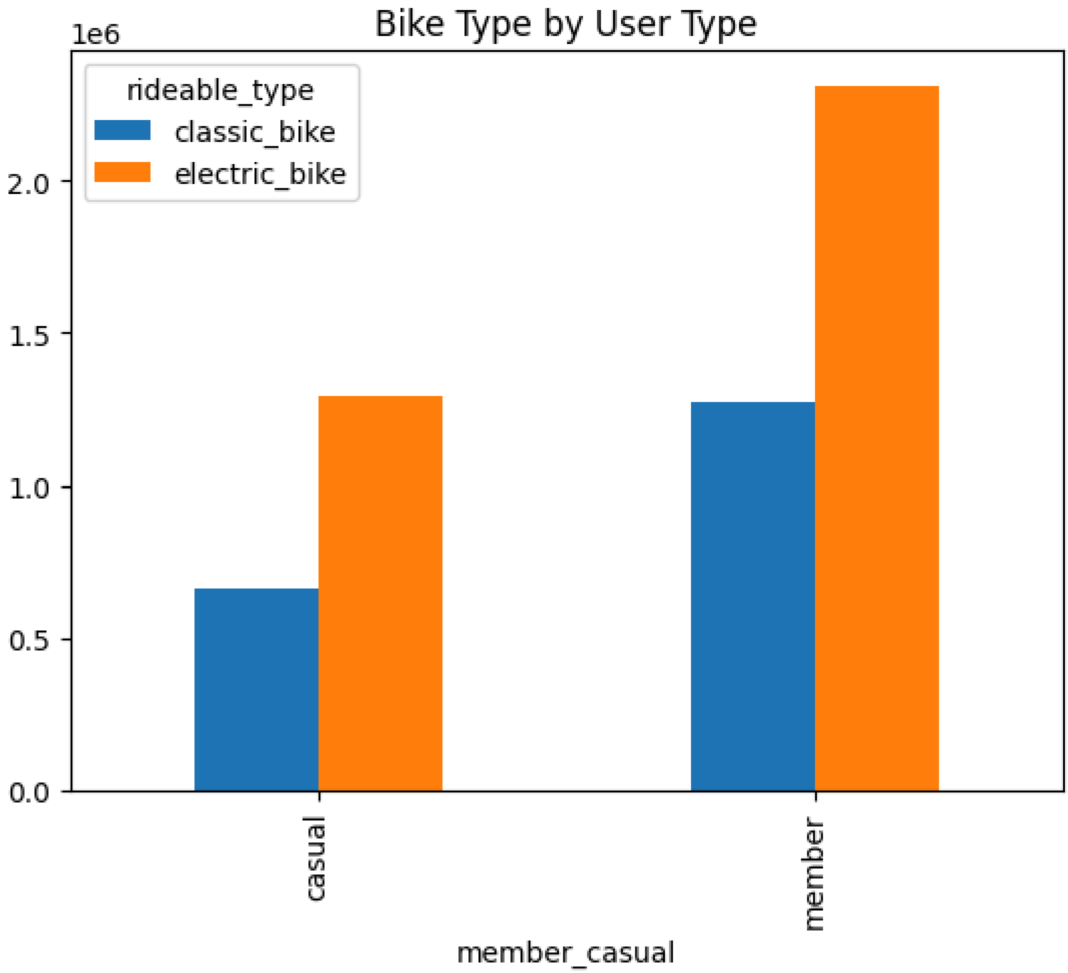

Member vs Casual Split
member
64.7%
count
35.3%
casual

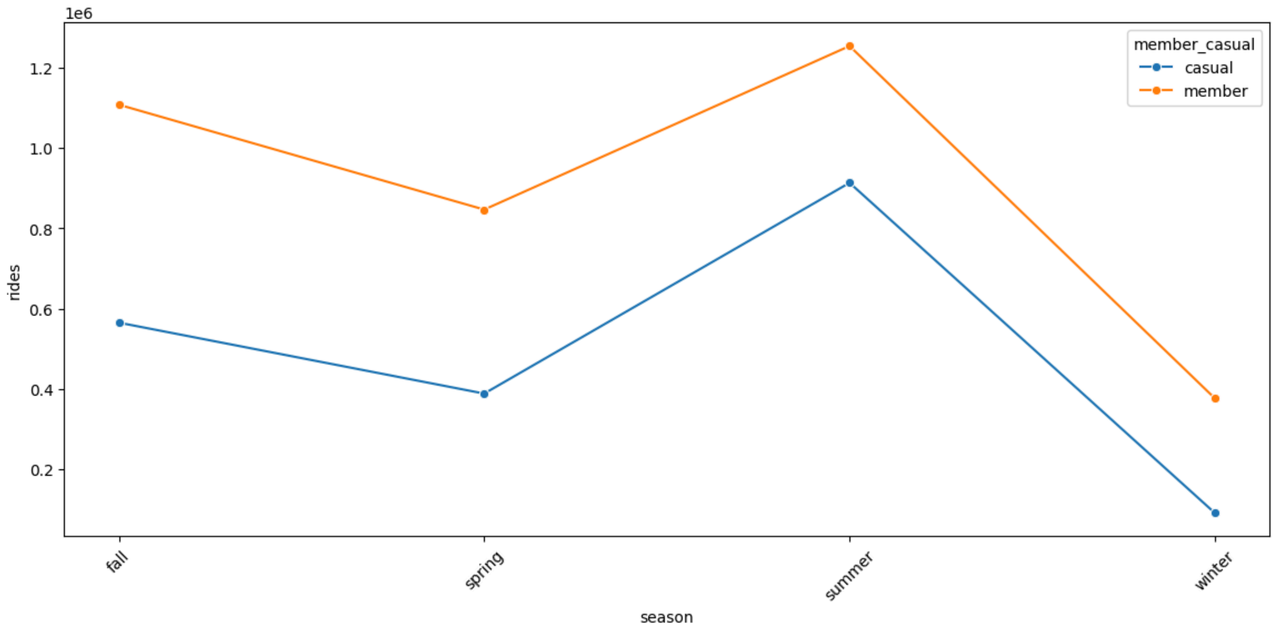

Average Ride Length by User Type
Minutes
0
5
10
15
20
casual
member
member_casual


day type prefernce
1e6
Ride Count
2.5
2.0
1.5
1.0
0.5
0.0
member_casual
casual
member
False
True
Day Type

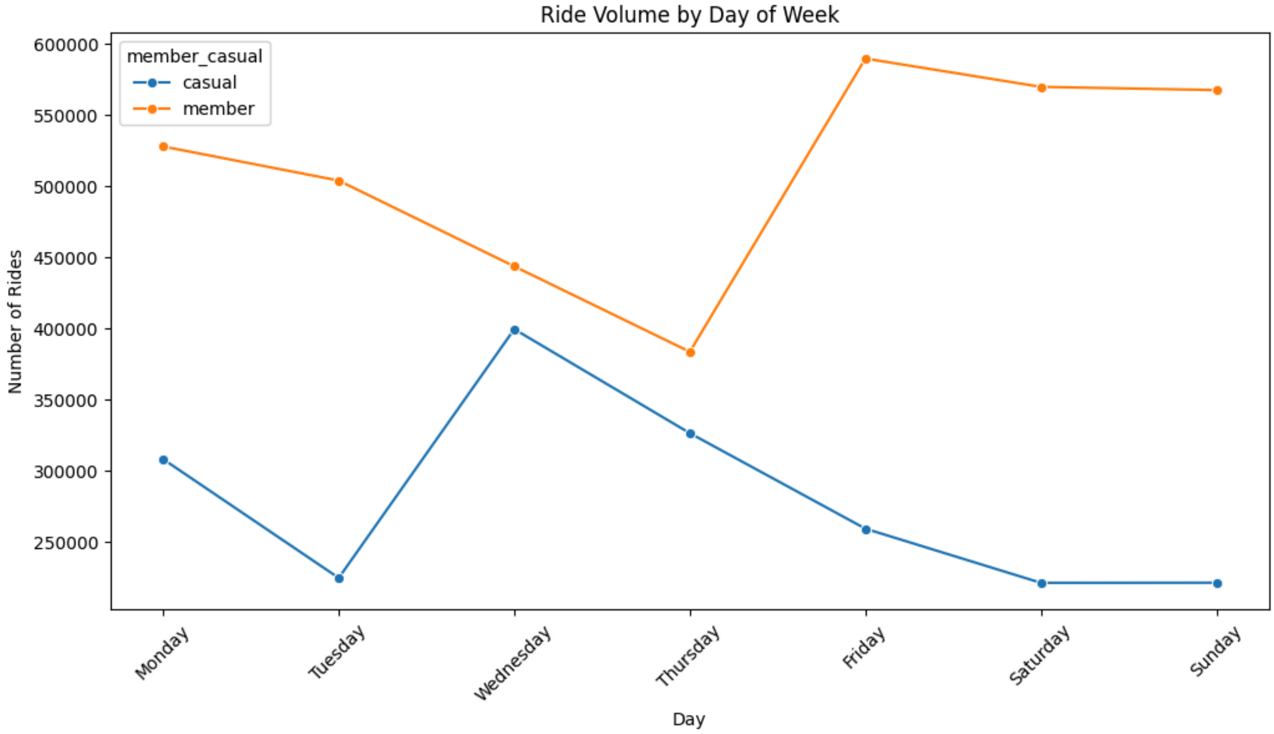

Hourly Usage Trends
Ride Count
400000
350000
300000
250000
200000
150000
100000
50000
0
member_casual
casual
member
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
hour

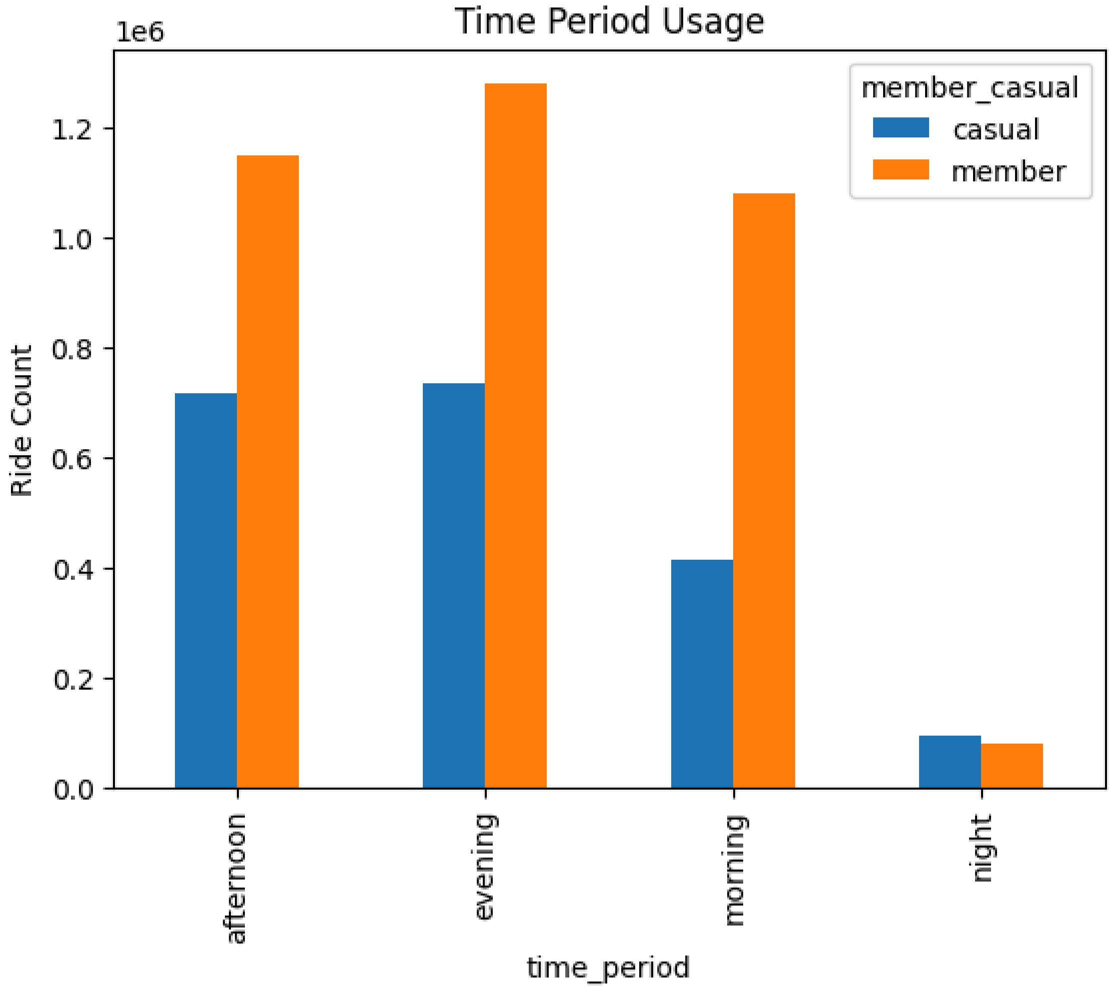

Monthly Ride Volume by Rider Type
member_casual
casual
member
Number of Rides
400000
300000
200000
100000
January
February
March
April
May
June
July
August
September
October
November
December
Month


Bike Type by User Type
1e6
rideable_type
classic_bike
electric_bike
2.0
1.5
1.0
0.5
0.0
casual
member
member_casual

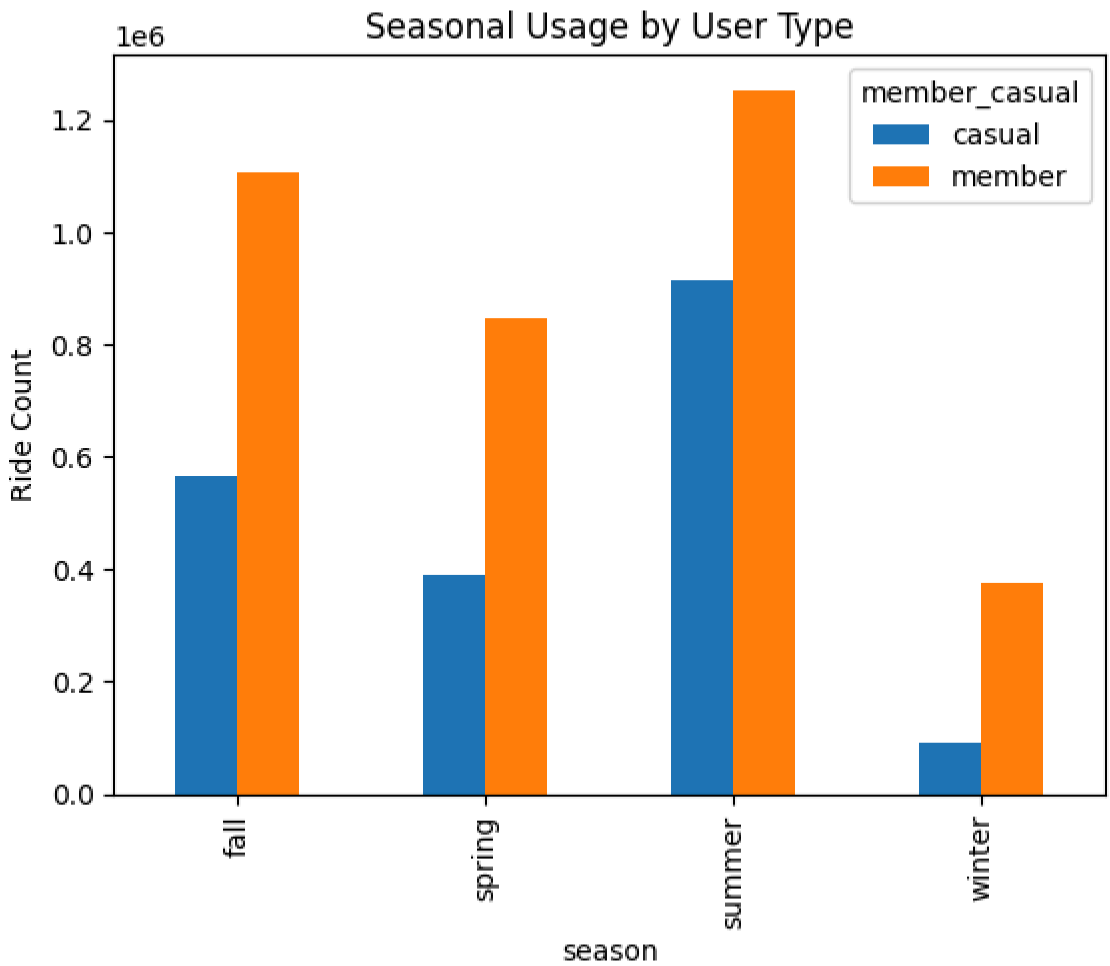


Ride Volume by Day of Week

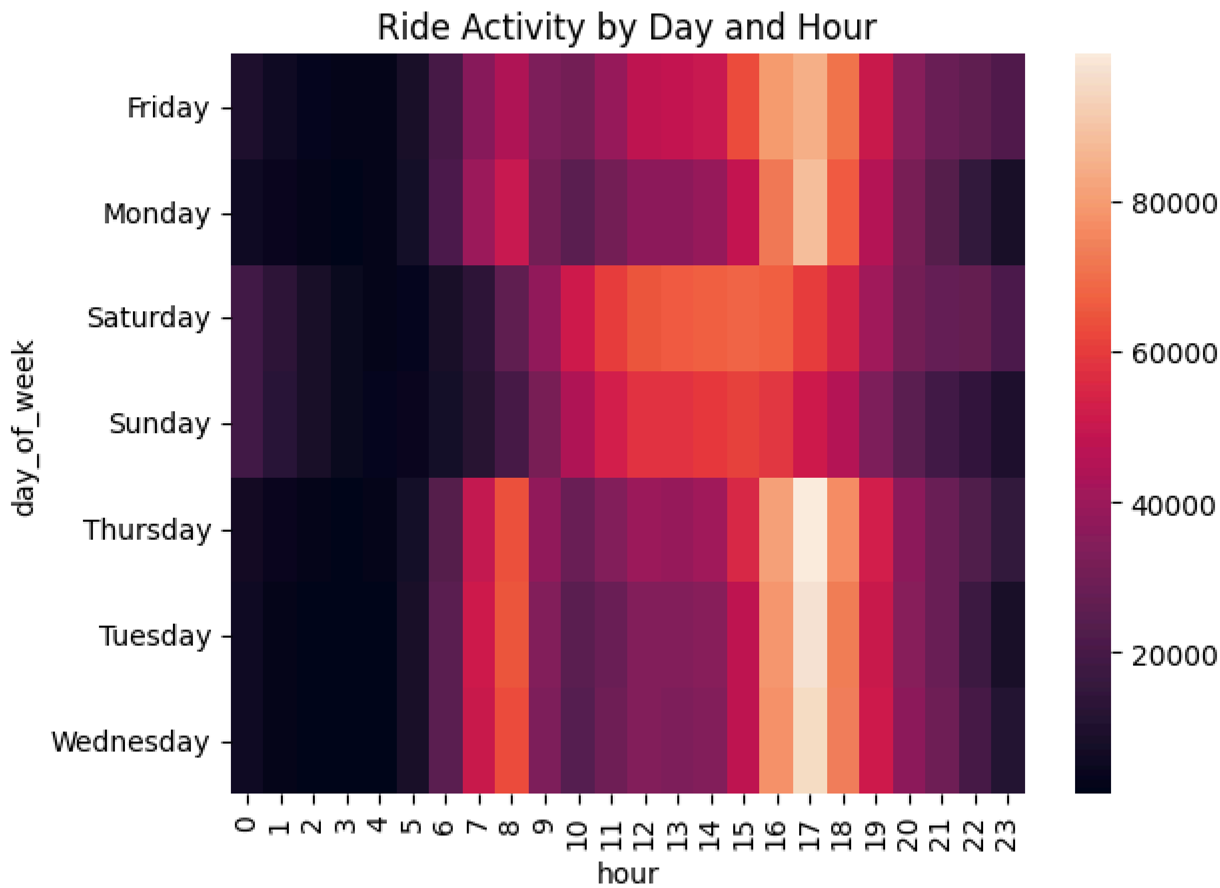




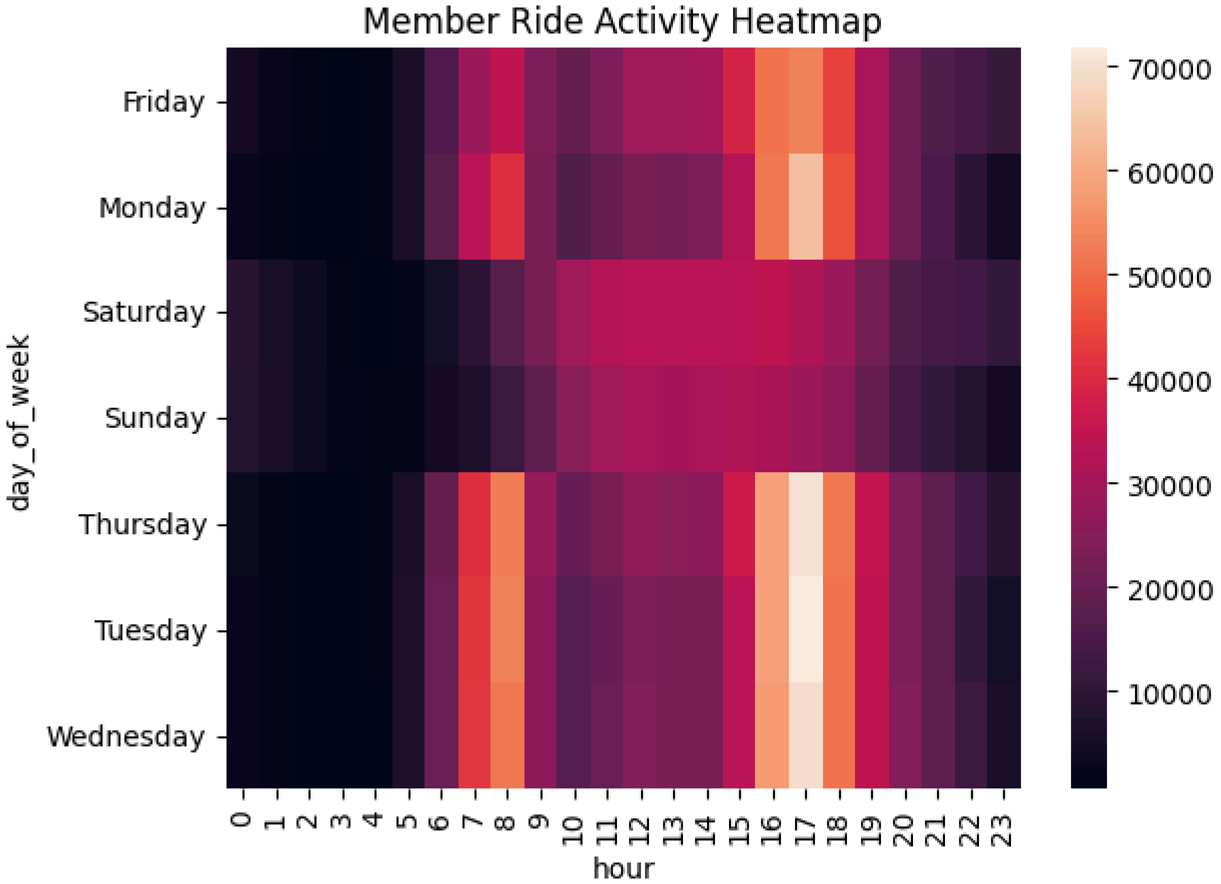
Time Period Usage
1e6
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Ride Count
member_casual
casual
member
afternoon
evening
morning
night
time_period

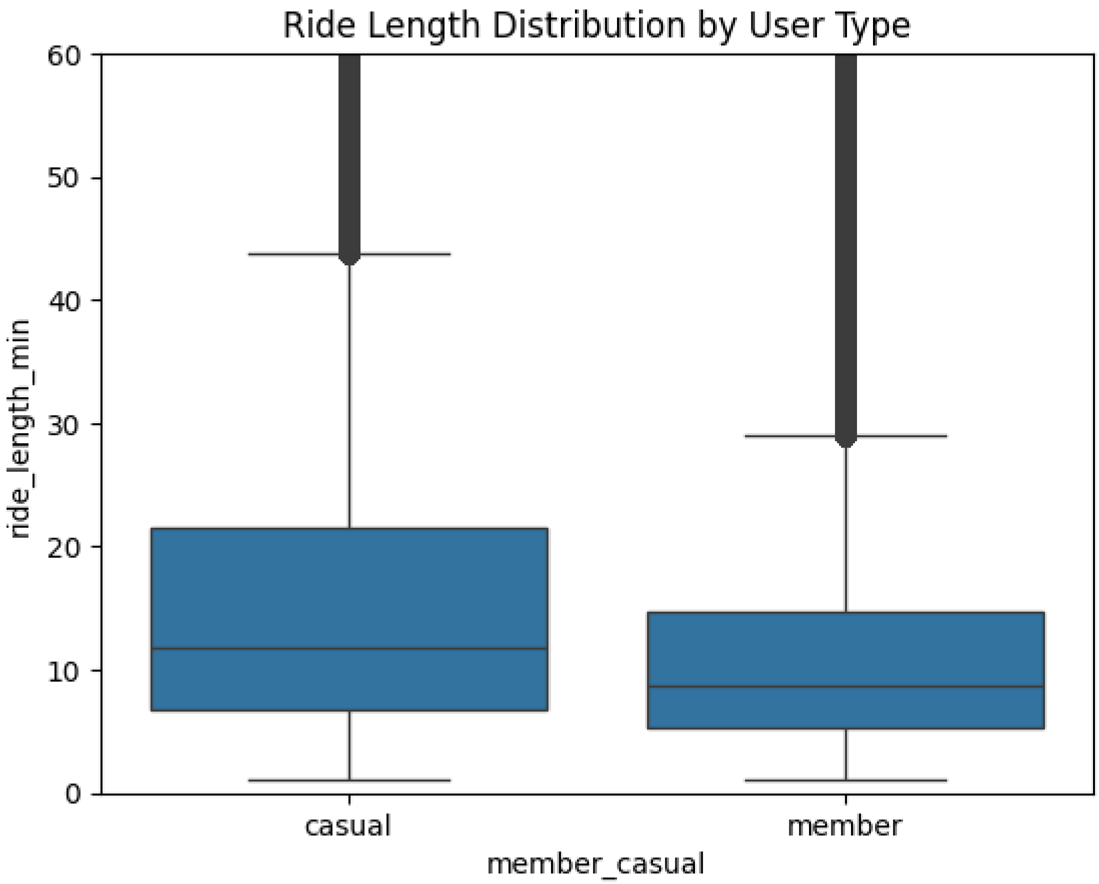


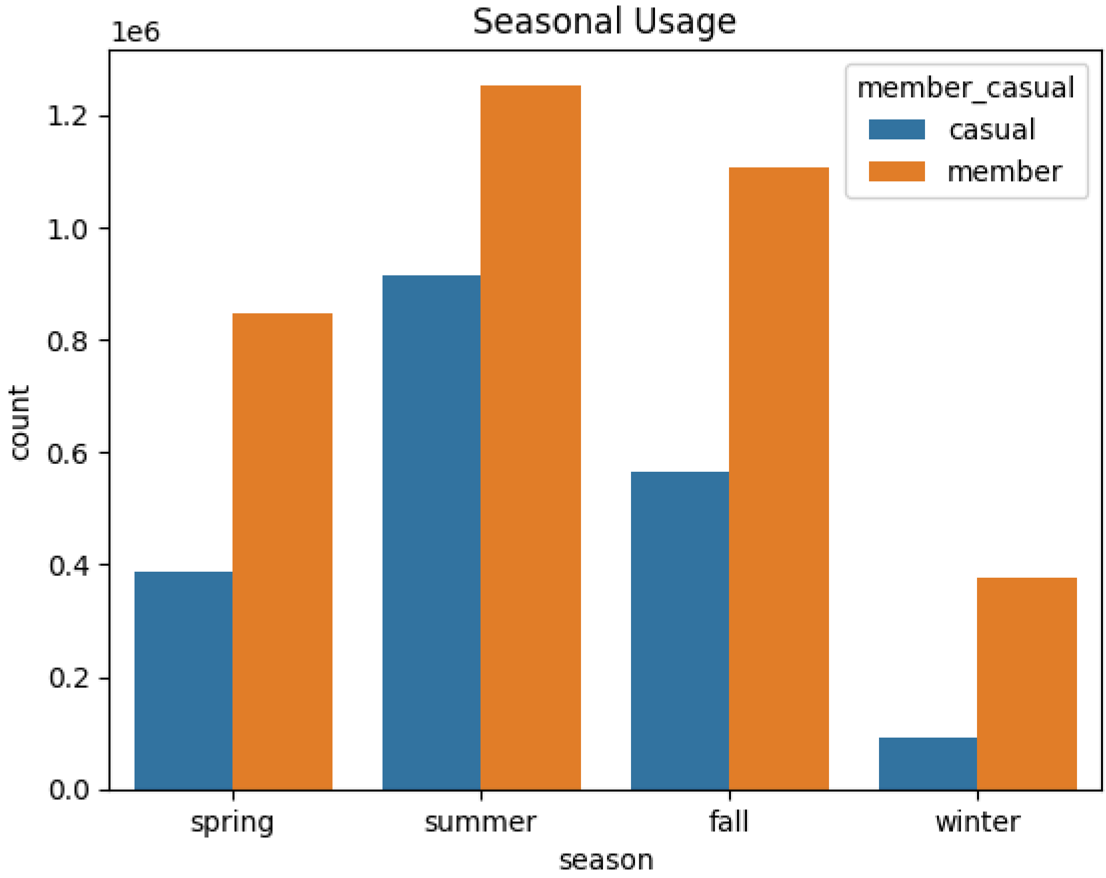
Seasonal Usage by User Type
1e6
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Ride Count
member_casual
casual
member
fall
spring
summer
winter
season

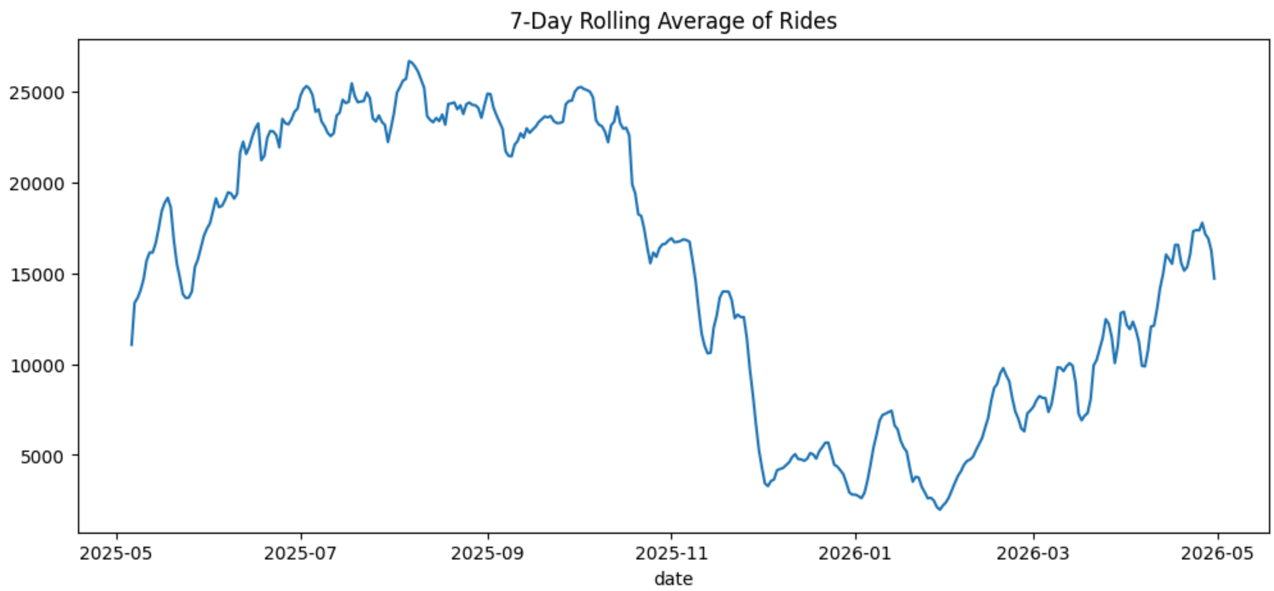


Ride Activity by Day and Hour
Friday
Monday
Saturday
Sunday
Thursday
Tuesday
Wednesday
day_of_week
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
hour
80000
60000
40000
20000


Member Ride Activity Heatmap
Friday
Monday
Saturday
Sunday
Thursday
Tuesday
Wednesday
day_of_week
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
hour
70000
60000
50000
40000
30000
20000
10000


Ride Length Distribution by User Type
0
10
20
30
40
50
60
ride_length_min
casual
member
member_casual

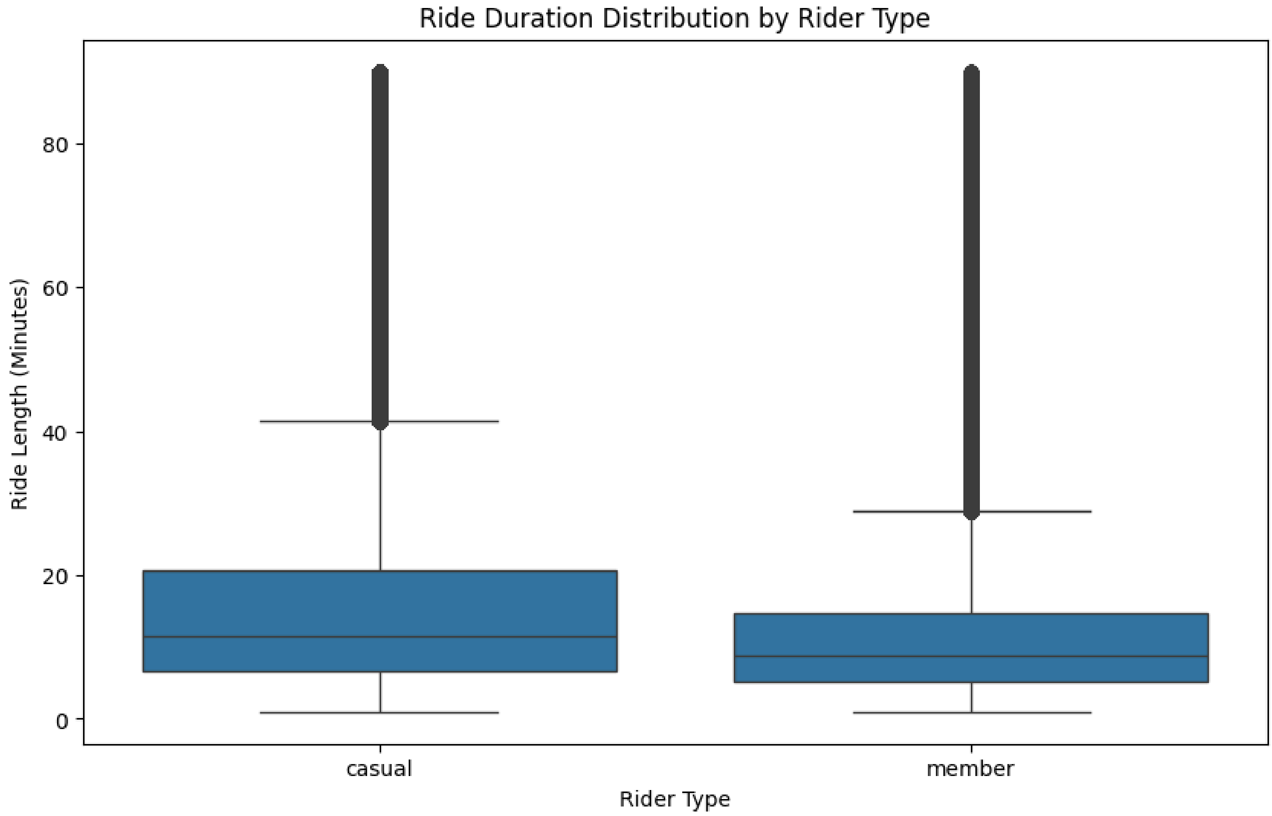




	count	mean	std	min	25%	50%	75%	max
member_casual								
casual	1957976.0	23.347497	82.935809	1.0	6.77	11.81	21.58	1574.9
member	3583847.0	12.690988	33.202348	1.0	5.22	8.75	14.74	1559.9

is_weekend	False	True
member_casual		
casual	1232778	725198
member	2757277	826570

time_period	afternoon	evening	morning	night
member_casual				
casual	714989	733213	415774	94000
member	1148257	1278250	1078247	79093



	count	mean	median	std	min	max
member_casual						
casual	1957976	23.35	11.81	82.94	1.0	1574.9
member	3583847	12.69	8.75	33.20	1.0	1559.9